

MATH 3260 - Chapter 1 Notes: Linear Systems

Feedback-Revised Student Notes

Kennesaw State University - MATH 3260

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Contents

Chapter overview	2
Section 1.1: Systems of Linear Equations	2
Learning objectives	2
1. Linear equations	2
2. Worked example: one solution	3
4. The three possible outcomes	3
5. One equation in three variables	4
6. Application: Spurs fan-shop bundles	4
7. Detailed preview: elimination in three variables	5
Section 1.1 terminology	7
Section 1.2: Gaussian Elimination and Gauss-Jordan Elimination	8
Learning objectives	8
1. Matrices as organized data	8
2. Elementary row operations	8
3. Row-echelon form and reduced row-echelon form	9
4. Gaussian elimination: full application example	9
5. Gauss-Jordan elimination: continue all the way to RREF	11
6. Gaussian versus Gauss-Jordan	12
7. Detecting an inconsistent system	12
8. Infinitely many solutions from RREF	12
9. Homogeneous systems	13
Section 1.2 terminology	14
Section 1.3: Applications of Linear Systems	15
Learning objectives	15
1. A modeling workflow	15
2. Curve fitting: a line through two points	15
3. Quadratic interpolation through three points	16
4. Network flow and conservation	17
5. Electrical networks and Kirchhoff's laws	18
6. Optional application: color-sensor calibration	19
Section 1.3 terminology	20

Chapter 1 wrap-up	20
Mathematical storyline	20

These notes are synchronized with the feedback-revised student notebook. They emphasize definitions, theorems, worked mathematics, conceptual connections, and interpretation while omitting notebook-specific code and worksheet instructions.

Chapter overview

1. **Section 1.1:** Systems of Linear Equations
2. **Section 1.2:** Gaussian Elimination and Gauss-Jordan Elimination
3. **Section 1.3:** Applications of Linear Systems
4. **Chapter summary and review**

Section 1.1: Systems of Linear Equations

Learning objectives

By the end of this section, students should be able to:

- recognize linear and nonlinear equations;
- explain what it means to solve a system;
- distinguish one solution, no solution, and infinitely many solutions;
- connect algebraic solutions with line and plane intersections;
- write parametric solution sets;
- classify systems as consistent or inconsistent;
- perform a first elimination calculation by hand;
- verify solutions with Python.

Why systems matter

A system is a collection of constraints that must hold simultaneously. The same structure appears in pricing, mixtures, traffic, circuits, computer graphics, data fitting, and resource allocation.

1. Linear equations

A linear equation in variables $x_1, x_2, \dots, x_n$ has the form

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = b,$$

where the coefficients $a_1, \dots, a_n$ and the constant b are fixed numbers. Variables occur only to the first power; variables are not multiplied together, placed in denominators, or used inside nonlinear functions.

Definition: system and solution

A **system of linear equations** is a collection of linear equations involving the same variables. A **solution** is an ordered list of values that satisfies every equation in the system at the same time.

Quick classification: linear or nonlinear?

Classify each example before checking the answer.

1. $4x - 3y = 21$
2. $x^2 + y = 9$
3. $2x_1 - x_2 + x_3 = 20$
4. $xy - z = 1$
5. $5u - 2v + 7w = 0$
6. $\sin x + y = 3$

2. Worked example: one solution

Solve

$$\begin{cases} u + v = 21, \\ u - v = -3. \end{cases}$$

Add the equations:

$$(u + v) + (u - v) = 21 + (-3),$$

so

$$2u = 18 \implies u = 9.$$

Substitute $u = 9$ into $u + v = 21$:

$$9 + v = 21 \implies v = 12.$$

Therefore,

$$(u, v) = (9, 12).$$

The numbers are Spurs-inspired, but the system itself is an original teaching example.

Verify rather than merely trust

A candidate solution must satisfy **both** equations.

4. The three possible outcomes

For a linear system in two variables, exactly one of the following occurs:

Geometry	Algebraic meaning	Number of solutions
Distinct intersecting lines	Two independent constraints	One
Distinct parallel lines	Contradictory constraints	None
Same line	One equation repeats the other	Infinitely many

Worked example: no solution

$$\begin{cases} 2x + y = 20, \\ 4x + 2y = 42. \end{cases}$$

Multiplying the first equation by 2 gives

$$4x + 2y = 40,$$

but the second equation requires $4x + 2y = 42$. The same expression cannot equal both 40 and 42. Therefore the system is inconsistent.

Why the contradiction settles the original system. Multiplying an equation by a nonzero constant is a legal, reversible operation, so it preserves the solution set. Therefore, when the valid algebra reveals

$$40 = 42,$$

the contradiction was already built into the original system. This is why the system is inconsistent.

Worked example: infinitely many solutions

$$\begin{cases} 3x - y = 9, \\ 6x - 2y = 18. \end{cases}$$

The second equation is twice the first, so both equations describe the same line. Let $x = t$. Then

$$3t - y = 9 \implies y = 3t - 9.$$

The complete solution set is

$$(x, y) = (t, 3t - 9), \quad t \in \mathbb{R}.$$

Parameter interpretation

A parameter does not mean “unknown forever.” It means the system allows a family of solutions. Choosing any real value of the parameter produces one member of that family.

5. One equation in three variables

Consider

$$x + 2y - z = 21.$$

There are three variables but only one constraint. Let

$$y = s, \quad z = t.$$

Then

$$x = 21 - 2s + t.$$

Therefore

$$(x, y, z) = (21 - 2s + t, s, t), \quad s, t \in \mathbb{R}.$$

Geometrically, the solution set is a plane in $\mathbb{R}^3$.

6. Application: Spurs fan-shop bundles

Let

- c = price of a cap;
- p = price of a poster.

Suppose two original promotional bundles are priced as follows:

$$\begin{cases} 2c + p = 30, \\ c + 2p = 33. \end{cases}$$

The numbers are chosen so the solution uses familiar Spurs-related values, but this is not actual store pricing.

Solve by elimination:

Multiply the second equation by 2:

$$2c + 4p = 66.$$

Subtract the first equation:

$$(2c + 4p) - (2c + p) = 66 - 30,$$

so

$$3p = 36 \implies p = 12.$$

Then

$$2c + 12 = 30 \implies c = 9.$$

Thus the modeled prices are $c = 9$ dollars and $p = 12$ dollars.

7. Detailed preview: elimination in three variables

Solve

$$\begin{cases} x + y + z = 31, \\ 2x - y + 3z = 36, \\ -x + 2y + z = -2. \end{cases}$$

Write the augmented matrix:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 31 \\ 2 & -1 & 3 & 36 \\ -1 & 2 & 1 & -2 \end{array} \right].$$

Where the augmented matrix comes from

Each equation becomes one row. The coefficients of x, y, z occupy the first three entries, and the right-hand-side constant is appended after the vertical bar. For example,

$$x + y + z = 31$$

becomes

$$[1, 1, 1 \mid 31].$$

Thus the system corresponds to

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 31 \\ 2 & -1 & 3 & 36 \\ -1 & 2 & 1 & -2 \end{array} \right].$$

Full-matrix view of the elimination

Rather than tracking only the changing row, it is useful to see the entire matrix at each stage:

$$\begin{aligned} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 31 \\ 2 & -1 & 3 & 36 \\ -1 & 2 & 1 & -2 \end{array} \right] &\longrightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & 31 \\ 0 & -3 & 1 & -26 \\ -1 & 2 & 1 & -2 \end{array} \right] \\ &\longrightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & 31 \\ 0 & -3 & 1 & -26 \\ 0 & 3 & 2 & 29 \end{array} \right] \longrightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & 31 \\ 0 & -3 & 1 & -26 \\ 0 & 0 & 3 & 3 \end{array} \right]. \end{aligned}$$

Eliminate x from Row 2 using

$$R_2 \leftarrow R_2 - 2R_1.$$

Entry-by-entry:

$$[2, -1, 3 \mid 36] - 2[1, 1, 1 \mid 31] = [0, -3, 1 \mid -26].$$

Eliminate x from Row 3 using

$$R_3 \leftarrow R_3 + R_1.$$

Entry-by-entry:

$$[-1, 2, 1 \mid -2] + [1, 1, 1 \mid 31] = [0, 3, 2 \mid 29].$$

Eliminate y from Row 3 using

$$R_3 \leftarrow R_3 - R_2.$$

$$[0, 3, 2 \mid 29] + [0, -3, 1 \mid -26] = [0, 0, 3 \mid 3].$$

Now back-substitute:

$$3z = 3 \implies z = 1.$$

From Row 2,

$$-3y + z = -26$$

so

$$-3y + 1 = -26 \implies -3y = -27 \implies y = 9.$$

From Row 1,

$$x + y + z = 31,$$

so

$$x + 9 + 1 = 31 \implies x = 21.$$

Therefore,

$$(x, y, z) = (21, 9, 1).$$

Common mistakes in Section 1.1

- Checking only one equation.
- Assuming every system has one solution.
- Giving one point when a parameterized family is required.
- Confusing a free variable with a pivot variable.
- Reporting a numerical answer without interpreting its meaning in context.

Section 1.1 terminology

Term	Meaning
Linear equation	An equation of the form $a_1x_1 + \cdots + a_nx_n = b$
System of linear equations	Several linear equations involving the same variables
Solution	Values satisfying every equation simultaneously
Solution set	The collection of all solutions
Consistent system	A system with at least one solution
Inconsistent system	A system with no solution
Unique solution	Exactly one solution
Free variable	A variable allowed to take arbitrary parameter values
Parameter	A symbol representing freely chosen values in a solution family
Parametric form	A description of all solutions using one or more parameters
Plane	The solution set of one nondegenerate linear equation in three variables

Section 1.2: Gaussian Elimination and Gauss-Jordan Elimination

Learning objectives

Students should be able to:

- identify matrix dimensions and entries;
- construct coefficient and augmented matrices;
- perform all three elementary row operations;
- distinguish row-echelon form from reduced row-echelon form;
- solve systems using Gaussian elimination and back-substitution;
- solve systems using Gauss-Jordan elimination;
- detect inconsistent and dependent systems from row-reduced matrices;
- analyze homogeneous systems;
- reproduce each matrix step manually and verify it with Python.

1. Matrices as organized data

An $m \times n$ matrix has m rows and n columns:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}.$$

The entry a_{ij} lies in row i and column j .

Coefficient and augmented matrices

For $A\mathbf{x} = \mathbf{b}$, the coefficient matrix contains only the variable coefficients. The augmented matrix $[A \mid \mathbf{b}]$ appends the constants as a final column.

For example,

$$\begin{cases} x + 2y - z = 21, \\ 3x - y + 4z = 20 \end{cases}$$

has coefficient matrix

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & -1 & 4 \end{bmatrix}$$

and augmented matrix

$$[A \mid b] = \left[\begin{array}{ccc|c} 1 & 2 & -1 & 21 \\ 3 & -1 & 4 & 20 \end{array} \right].$$

2. Elementary row operations

Each legal row operation preserves the solution set.

1. **Row replacement:** $R_i \leftarrow R_i + kR_j$.
2. **Row interchange:** $R_i \leftrightarrow R_j$.
3. **Row scaling:** $R_i \leftarrow kR_i$, where $k \neq 0$.

Why row operations are valid

Replacing one equation by a nonzero multiple, swapping equation order, or adding a multiple of one equation to another does not change which variable values satisfy the whole system.

3. Row-echelon form and reduced row-echelon form

A matrix is in **row-echelon form (REF)** when:

1. all nonzero rows are above zero rows;
2. each leading entry is to the right of the leading entry above it;
3. all entries below a leading entry are zero.

It is in **reduced row-echelon form (RREF)** when, additionally:

4. each leading entry is 1;
5. each pivot 1 is the only nonzero entry in its column.

Visual cue for leading entries

In REF, identify the first nonzero entry in each nonzero row. These entries form the staircase. In RREF they are scaled to isolated pivot 1s:

$$\begin{bmatrix} \boxed{1} & 2 & -1 & 4 \\ 0 & \boxed{3} & 5 & 7 \\ 0 & 0 & \boxed{2} & 6 \end{bmatrix}, \quad \begin{bmatrix} \boxed{1} & 0 & 0 & 9 \\ 0 & \boxed{1} & 0 & 20 \\ 0 & 0 & \boxed{1} & 21 \end{bmatrix}.$$

4. Gaussian elimination: full application example

A fictional Spurs fan shop sells three items:

- c = cap price;
- s = shirt price;
- p = poster price.

Three original bundles produce the system

$$\begin{cases} c + s + p = 50, \\ 2c + s + 3p = 101, \\ c + 2s + 4p = 133. \end{cases}$$

The goal is not the realism of the bundle design; the goal is to model three unknown quantities and execute every elimination step.

Step 0: write the augmented matrix

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 50 \\ 2 & 1 & 3 & 101 \\ 1 & 2 & 4 & 133 \end{array} \right].$$

Step 1: eliminate the first entry of Row 2

Use

$$R_2 \leftarrow R_2 - 2R_1.$$

Compute every entry:

$$[2, 1, 3 \mid 101] - 2[1, 1, 1 \mid 50] = [0, -1, 1 \mid 1].$$

Step 2: eliminate the first entry of Row 3

Use

$$R_3 \leftarrow R_3 - R_1.$$

$$[1, 2, 4 \mid 133] - [1, 1, 1 \mid 50] = [0, 1, 3 \mid 83].$$

Step 3: eliminate the second entry of Row 3

Row 2 has -1 in the pivot position and Row 3 has 1. Add Row 2 to Row 3:

$$R_3 \leftarrow R_3 + R_2.$$

$$[0, 1, 3 \mid 83] + [0, -1, 1 \mid 1] = [0, 0, 4 \mid 84].$$

The matrix is now in row-echelon form. Translate each row back into an equation:

$$\begin{aligned} c + s + p &= 50, \\ -s + p &= 1, \\ 4p &= 84. \end{aligned}$$

Back-substitution

From Row 3:

$$p = \frac{84}{4} = 21.$$

From Row 2:

$$-s + p = 1$$

so

$$-s + 21 = 1 \implies -s = -20 \implies s = 20.$$

From Row 1:

$$c + 20 + 21 = 50 \implies c = 9.$$

Thus

$$(c, s, p) = (9, 20, 21).$$

5. Gauss-Jordan elimination: continue all the way to RREF

Solve

$$\begin{cases} x + y + z = 50, \\ x + 2y + z = 71, \\ 2x + y + 3z = 88. \end{cases}$$

We will reduce the augmented matrix until every pivot column has zeros above and below its pivot.

Forward elimination

First,

$$R_2 \leftarrow R_2 - R_1,$$

which gives

$$[1, 2, 1 \mid 71] - [1, 1, 1 \mid 50] = [0, 1, 0 \mid 21].$$

Next,

$$R_3 \leftarrow R_3 - 2R_1,$$

so

$$[2, 1, 3 \mid 88] - 2[1, 1, 1 \mid 50] = [0, -1, 1 \mid -12].$$

Now eliminate the -1 below the second pivot:

$$R_3 \leftarrow R_3 + R_2,$$

so

$$[0, -1, 1 \mid -12] + [0, 1, 0 \mid 21] = [0, 0, 1 \mid 9].$$

Backward elimination to RREF

Eliminate z from Row 1:

$$R_1 \leftarrow R_1 - R_3.$$

Eliminate y from Row 1:

$$R_1 \leftarrow R_1 - R_2.$$

The RREF is

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 20 \\ 0 & 1 & 0 & 21 \\ 0 & 0 & 1 & 9 \end{array} \right],$$

so

$$(x, y, z) = (20, 21, 9).$$

6. Gaussian versus Gauss-Jordan

Method	Stopping point	Final work
Gaussian elimination	Row-echelon form	Back-substitution
Gauss-Jordan elimination	Reduced row-echelon form	Read solution directly

Gaussian elimination often uses fewer operations. Gauss-Jordan produces a more informative final matrix.

7. Detecting an inconsistent system

Consider

$$\begin{cases} x + 2y = 21, \\ 2x + 4y = 43. \end{cases}$$

Start with

$$\left[\begin{array}{cc|c} 1 & 2 & 21 \\ 2 & 4 & 43 \end{array} \right].$$

Use

$$R_2 \leftarrow R_2 - 2R_1.$$

Then

$$[2, 4 \mid 43] - 2[1, 2 \mid 21] = [0, 0 \mid 1].$$

The last row means $0 = 1$, which is impossible.

Because elementary row operations are reversible and preserve the solution set, the row

$$[0 \quad 0 \mid 1]$$

does more than show that the *reduced* system is impossible. It proves that the original system had no solution either.

8. Infinitely many solutions from RREF

Consider

$$\begin{cases} x + y + z = 21, \\ 2x + 2y + 2z = 42, \\ x - y + z = 1. \end{cases}$$

The augmented matrix is

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 21 \\ 2 & 2 & 2 & 42 \\ 1 & -1 & 1 & 1 \end{array} \right].$$

The RREF equations are

$$x + z = 11, \quad y = 10.$$

Let $z = t$. Then

$$x = 11 - t,$$

so

$$(x, y, z) = (11 - t, 10, t), \quad t \in \mathbb{R}.$$

9. Homogeneous systems

A homogeneous system has the form

$$A\mathbf{x} = \mathbf{0}.$$

It always has the **trivial solution** $\mathbf{x} = \mathbf{0}$. It may also have nontrivial solutions.

Consider

$$\begin{cases} x + y - z = 0, \\ 2x + 2y - 2z = 0, \\ x - y + z = 0. \end{cases}$$

The RREF equations are

$$x = 0, \quad y - z = 0.$$

Let $z = t$. Then $y = t$, and

$$(x, y, z) = (0, t, t) = t(0, 1, 1).$$

The trivial solution occurs when $t = 0$; every $t \neq 0$ gives a nontrivial solution.

Common mistakes in Section 1.2

- Changing only part of a row instead of the entire row.
- Forgetting to apply a row operation to the augmented column.
- Using a newly changed row inconsistently within the same step.
- Stopping at REF and reading the answer without back-substitution.
- Calling a zero row a contradiction; only a row of the form $[0 \ \cdots \ 0 \mid c]$ with $c \neq 0$ is contradictory.
- Assuming a homogeneous system has only the trivial solution.

Section 1.2 terminology

Term	Meaning
Matrix	Rectangular array of entries
$m \times n$ matrix	Matrix with m rows and n columns
Entry a_{ij}	Entry in row i , column j
Coefficient matrix	Matrix containing variable coefficients
Augmented matrix	Coefficient matrix with the constants column appended
Elementary row operation	Row replacement, row interchange, or nonzero row scaling
Row equivalent	Related by a sequence of elementary row operations
Leading entry	First nonzero entry in a nonzero row
Pivot position	Position of a leading entry in an echelon form
Row-echelon form	Stair-step form with zeros below pivots
Reduced row-echelon form	Echelon form with pivot 1s isolated in their columns
Gaussian elimination	Reduce to REF and back-substitute
Gauss-Jordan elimination	Reduce completely to RREF
Homogeneous system	A system $A\mathbf{x} = \mathbf{0}$
Trivial solution	The zero solution of a homogeneous system
Nontrivial solution	Any nonzero solution of a homogeneous system

Section 1.3: Applications of Linear Systems

Learning objectives

Students should be able to:

- translate an applied situation into variables, equations, and a matrix model;
- fit lines and quadratic polynomials through data points;
- solve and interpret network-flow systems;
- apply conservation laws and physical restrictions;
- formulate simple circuit systems using Kirchhoff's laws;
- use Python to visualize and explore how parameter changes affect a model.

1. A modeling workflow

A reliable applied-linear-algebra workflow is:

1. Define variables with units.
2. Translate each constraint into an equation.
3. Assemble $A\mathbf{x} = \mathbf{b}$.
4. Solve or parameterize the system.
5. Check mathematical and physical constraints.
6. Interpret the result in the original context.

Modeling warning

A mathematically correct solution can still be physically impossible. Negative traffic flow, negative product quantities, or implausible current directions require interpretation rather than automatic rejection.

2. Curve fitting: a line through two points

Suppose an original teaching dataset contains the points

$$(1, 21), \quad (3, 9).$$

We seek a line

$$y = mx + b.$$

Substituting the two points gives

$$\begin{cases} m + b = 21, \\ 3m + b = 9. \end{cases}$$

Subtract the first equation from the second:

$$2m = -12 \implies m = -6.$$

Then

$$-6 + b = 21 \implies b = 27.$$

Thus

$$y = -6x + 27.$$

3. Quadratic interpolation through three points

Fit

$$p(x) = ax^2 + bx + c$$

through the original, jersey-number-inspired points

$$(0, 9), \quad (1, 20), \quad (2, 21).$$

Substitute each point:

- At $x = 0$: $c = 9$.
- At $x = 1$: $a + b + c = 20$.
- At $x = 2$: $4a + 2b + c = 21$.

The coefficient system is

$$\begin{cases} c = 9, \\ a + b + c = 20, \\ 4a + 2b + c = 21. \end{cases}$$

In matrix form,

$$\left[\begin{array}{ccc|c} 0 & 0 & 1 & 9 \\ 1 & 1 & 1 & 20 \\ 4 & 2 & 1 & 21 \end{array} \right].$$

Use $R_2 \leftarrow R_2 - R_1$ and $R_3 \leftarrow R_3 - R_1$:

$$\left[\begin{array}{ccc|c} 0 & 0 & 1 & 9 \\ 1 & 1 & 0 & 11 \\ 4 & 2 & 0 & 12 \end{array} \right].$$

Now eliminate a from Row 3 using

$$R_3 \leftarrow R_3 - 4R_2 :$$

$$[4, 2, 0 \mid 12] - 4[1, 1, 0 \mid 11] = [0, -2, 0 \mid -32].$$

Thus

$$-2b = -32 \implies b = 16.$$

From $a + b = 11$,

$$a = 11 - 16 = -5.$$

Since $c = 9$,

$$p(x) = -5x^2 + 16x + 9.$$

4. Network flow and conservation

At every intersection, conservation requires

$$\text{total inflow} = \text{total outflow}.$$

Consider the original traffic network below. The values are vehicles per hour.

The node equations are chosen as follows:

- Node A : $x_1 + x_2 = 67$.
- Node B : $x_1 + 20 = x_3$.
- Node C : $x_2 = x_4 + 21$.
- Node D : $x_3 + x_4 = 66$. This equation is dependent on the first three balances.

Use the first three equations to expose the free variable. Let

$$x_1 = t.$$

Then

$$x_2 = 67 - t,$$

and

$$x_3 = t + 20.$$

Substitute into the Node C equation:

$$67 - t = x_4 + 21.$$

Therefore

$$x_4 = 46 - t.$$

The Node D balance is then automatic:

$$(t + 20) + (46 - t) = 66.$$

Thus

$$(x_1, x_2, x_3, x_4) = (t, 67 - t, t + 20, 46 - t).$$

For all flows to be nonnegative,

$$0 \leq t \leq 46.$$

Interpretation

The system does not determine a unique internal traffic pattern. It determines a one-parameter family. Operational restrictions, lane capacities, or measured flows would be needed to choose a specific value of t .

Since $x_1 = t$, the parameter has a concrete traffic meaning. For example,

$$t = 21 \implies (x_1, x_2, x_3, x_4) = (21, 46, 41, 25).$$

So this feasible pattern sends 21, 46, 41, and 25 vehicles per hour through the four internal roads. The bound $0 \leq t \leq 46$ keeps every modeled flow nonnegative.

5. Electrical networks and Kirchhoff's laws

For a simple circuit model:

- **Kirchhoff's Current Law:** current entering a junction equals current leaving it.
- **Kirchhoff's Voltage Law:** the signed voltage changes around a closed loop sum to zero.
- **Ohm's law:** $V = RI$.

Suppose the model produces

$$\begin{cases} I_1 - I_2 + I_3 = 0, \\ 2I_1 + I_2 = 38, \\ I_2 + 3I_3 = 53. \end{cases}$$

The first equation is a junction balance. The last two are loop equations.

Use Row 1 as the first pivot row. Eliminate I_1 from Row 2:

$$R_2 \leftarrow R_2 - 2R_1.$$

$$[2, 1, 0 \mid 38] - 2[1, -1, 1 \mid 0] = [0, 3, -2 \mid 38].$$

Now eliminate I_2 from Row 3. A fraction-free choice is

$$R_3 \leftarrow 3R_3 - R_2.$$

First,

$$3R_3 = [0, 3, 9 \mid 159].$$

Then

$$[0, 3, 9 \mid 159] - [0, 3, -2 \mid 38] = [0, 0, 11 \mid 121].$$

Back-substitute:

$$11I_3 = 121 \implies I_3 = 11.$$

From Row 2,

$$3I_2 - 2I_3 = 38,$$

so

$$3I_2 - 22 = 38 \implies 3I_2 = 60 \implies I_2 = 20.$$

From Row 1,

$$I_1 - I_2 + I_3 = 0,$$

so

$$I_1 - 20 + 11 = 0 \implies I_1 = 9.$$

Therefore,

$$(I_1, I_2, I_3) = (9, 20, 11).$$

Interpretation

The solution gives the three modeled branch currents:

$$I_1 = 9, \quad I_2 = 20, \quad I_3 = 11.$$

Check:

$$9 - 20 + 11 = 0, \quad 2(9) + 20 = 38, \quad 20 + 3(11) = 53.$$

Thus the currents satisfy every equation in the circuit model.

6. Optional application: color-sensor calibration

A camera sensor combines unknown red, green, and blue response weights (r, g, b) . Three calibration cards produce

$$\begin{cases} r + g + b = 30, \\ 2r + g + 3b = 41, \\ r + 4g + 2b = 91. \end{cases}$$

Application discussion

Ask students what assumptions make each model linear. For curve fitting, coefficients enter linearly. For traffic, conservation uses sums and differences. For circuits with fixed resistances, Ohm's law and Kirchhoff's laws produce linear equations.

Common mistakes in Section 1.3

- Failing to define variables and units.
- Using data coordinates in the wrong polynomial order.
- Writing inflow minus outflow inconsistently from node to node.
- Ignoring nonnegativity or capacity restrictions.
- Treating a free parameter as a failure rather than meaningful flexibility.

- Reporting Python output without checking it in the original equations.

Section 1.3 terminology

Term	Meaning
Mathematical model	Mathematical representation of a real or designed situation
Interpolation	Constructing a function that passes exactly through specified data points
Polynomial coefficients	Unknown constants $a, b, c, \dots$ in a polynomial model
Vandermonde-type system	Linear system created by substituting data points into a polynomial
Network	Collection of nodes connected by directed or undirected links
Node	Junction at which flow is conserved
Flow conservation	Total inflow equals total outflow
Free-flow parameter	Parameter representing an undetermined internal flow
Feasible solution	Mathematically valid solution satisfying physical restrictions
Kirchhoff's Current Law	Current entering a junction equals current leaving it
Kirchhoff's Voltage Law	Signed voltage changes around a closed loop sum to zero
Ohm's law	Relationship $V = RI$
Sensitivity	How model outputs change when inputs or parameters change

Chapter 1 wrap-up

Mathematical storyline

- Linear equations encode constraints.
- A system asks for values satisfying all constraints simultaneously.
- Geometry explains the three solution possibilities.
- Matrices organize coefficients and constants.
- Row operations preserve the solution set.
- Gaussian elimination reaches REF and uses back-substitution.
- Gauss-Jordan elimination reaches RREF.
- Parameters describe solution families.
- Applications require interpretation and feasibility checks.